



SIMATS Online Programming Lab

C PROGRAMMING



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CO4 - AT1 - LEVEL 2 Exam

Level Exam

C

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Result Available

Your score: **84 / 100**

QUESTIONS

LEVEL 2 — MEDIUM

Q1

1DArray

Q2

structure

Q3

string

Q4

2DArray

4 / 4 saved

Level 2 — Medium **merge (M)** — 1DArray

Write a program to merge all overlapping intervals in an array.

Sample Input:

Enter number of intervals: 4

Enter intervals: [1, 3], [2, 6], [8, 10], [15, 18]

Sample Output:

Merged intervals: [1, 6], [8, 10], [15, 18]

Your Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct {
5     int start;
6     int end;
7 } Interval;
8
9
10 int compareIntervals(const void *a, const
11     Interval *i1 = (Interval *)a;
12     Interval *i2 = (Interval *)b;
```

```
13     if (i1->start != i2->start) {
14         return i1->start - i2->start;
15     }
16     return i1->end - i2->end;
17 }
18
19 int main() {
20     int n;
21
22     printf("\nEnter number of intervals:
23     if (scanf("%d", &n) != 1) return 0;
24
25     Interval *intervals = (Interval *)mal
26
27     for (int i = 0; i < n; i++) {
28         printf("\nEnter interval %d start
29
30         if (i == 0) {
31
32             scanf(" [%d,%d]", &intervals[
33         } else {
34
35             scanf(", [%d,%d]", &intervals[
36         }
37     }
38
39
40     qsort(intervals, n, sizeof(Interval),
41
42
43     int mergedCount = 0;
44     if (n > 0) {
45         int currentIdx = 0;
46         for (int i = 1; i < n; i++) {
47             if (intervals[i].start <= int
48
49                 if (intervals[i].end > ir
50                     intervals[currentIdx]
51                 }
52             } else {
```

```
53
54         currentIdx++;
55         intervals[currentIdx] = i
56     }
57 }
58 mergedCount = currentIdx + 1;
59 }
60
61
62 printf("\nMerged intervals: ");
63 for (int i = 0; i < mergedCount; i++)
64     printf("[%d, %d]s", intervals[i]
65 }
66 printf("\n");
67
68 free(intervals);
69 return 0;
70 }
71
```



Program Input (stdin)

Enter input if required...

Output

```
Enter number of intervals:
Enter interval 1 start and end:
Enter interval 2 start and end:
Enter interval 3 start and end:
Enter interval 4 start and end:
```



Run



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